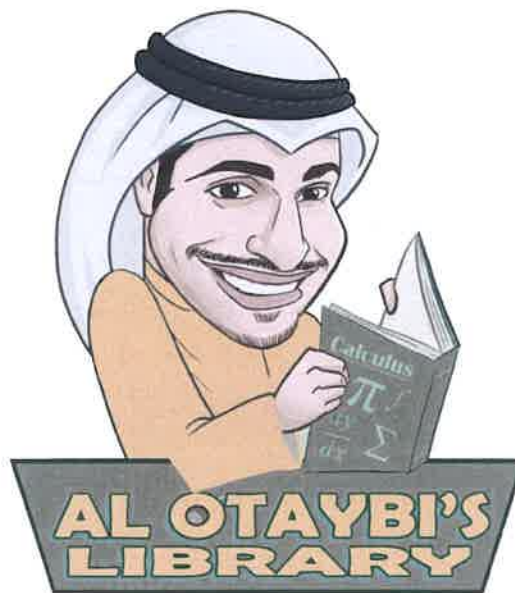


Kuwait University

2021/2022

Spring course



Calculus B

Second

Section : **11.1 (Part a/d)**

Review

11.1 Sequences.

(8 a)

مراجعة على اللمت



عند وجود دوال مثلثية



عند فشل التعويض

$\sin(\infty)$ or $\cos(\infty)$



Sandwich Theorem

(1) اطبق أحد القانونين

$$-1 \leq \frac{\sin^n \theta}{\cos^n \theta} \leq 1$$

الأس فردي

$$0 \leq \frac{\sin^n \theta}{\cos^n \theta} \leq 1$$

الأس زوجي

(2) أوصل الي بالنص للسؤال

(3) أخذ اللمت للأطراف

$\sin(0)$ or $\tan(0)$



$$\lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1$$

$$\lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1$$

$$\lim_{n \rightarrow \infty} \frac{1 - \cos\left(\frac{1}{n}\right)}{\frac{1}{n}} = 0$$

(1) الزاوية = المقام

(2) اللمت يسعى لصفر الزاوية

Evaluate the limit if it exist

$$\lim_{n \rightarrow \infty} \frac{\sin(n)}{n}$$

Sin n
 ∞ Sandwich

$\div n$

$$-1 \leq \sin n \leq 1$$

$$\frac{-1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{-1}{n} = -\frac{1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = \frac{1}{\infty} = 0$$

$\therefore$ By Sandwich Theorem $\implies \lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0$

$$\lim_{n \rightarrow \infty} \frac{\cos^2(n)}{\ln n}$$

$$\frac{\cos^2 \infty}{\ln \infty}$$

$\div \ln n$

$$0 \leq \cos^2 n \leq 1$$

$$0 \leq \frac{\cos^2 n}{\ln n} \leq \frac{1}{\ln n}$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{\ln n} = \frac{1}{\infty} = 0$$

$\therefore$ By Sandwich Theorem

$$\lim_{n \rightarrow \infty} \frac{\cos^2 n}{\ln n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1 + \sin(n)}{\sqrt{n}}$$

$$\frac{1 + \sin \infty}{\infty}$$

$$\lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{n}} + \frac{\sin n}{\sqrt{n}} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} + \lim_{n \rightarrow \infty} \frac{\sin n}{\sqrt{n}}$$

$$= \frac{1}{\infty}$$

$$= 0$$

$$-1 \leq \sin n \leq 1$$

$$\div \sqrt{n}$$

$$\frac{-1}{\sqrt{n}} \leq \frac{\sin n}{\sqrt{n}} \leq \frac{1}{\sqrt{n}}$$

$$\lim_{n \rightarrow \infty} \frac{-1}{\sqrt{n}} = \frac{-1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = \frac{1}{\infty} = 0$$

∴ By Sandwich Theorem

$$\lim_{n \rightarrow \infty} \frac{\sin n}{\sqrt{n}} = 0$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} \frac{1 + \sin n}{\sqrt{n}} = 0 + 0 = 0$$

$$\lim_{n \rightarrow \infty} \left(\frac{\sqrt{1 + \tan^{-1} n} - \sqrt{1 + \sin n}}{n^2} \right)$$

$$\lim_{n \rightarrow \infty} \left[\frac{\sqrt{1 + \tan^{-1} n}}{n^2} - \frac{\sqrt{1 + \sin n}}{n^2} \right]$$

$$\lim_{n \rightarrow \infty} \frac{\sqrt{1 + \tan^{-1} n}}{n^2} - \lim_{n \rightarrow \infty} \frac{\sqrt{1 + \sin n}}{n^2}$$

$$\frac{\sqrt{1 + \sin n}}{\infty}$$

$$= \frac{\sqrt{1 + \tan^{-1} n}}{\infty} - 1 \leq \sin n \leq 1$$

$$= \frac{\sqrt{1 + \frac{\pi}{2}}}{\infty}$$

$$= 0$$

$$0 \leq 1 + \sin n \leq 2$$

$$\div \sqrt{n^2} \Rightarrow \frac{\sqrt{1 + \sin n}}{n^2} \leq \frac{\sqrt{2}}{n^2}$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{\sqrt{2}}{n^2} = \frac{\sqrt{2}}{\infty} = 0$$

By Sandwich Theorem

$$\lim_{n \rightarrow \infty} \frac{\sqrt{1 + \sin n}}{n^2} = 0$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} \frac{\sqrt{1 + \tan^{-1} n} - \sqrt{1 + \sin n}}{n^2}$$

$$= 0 - 0 = 0$$

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{2}{n}\right)$$

$$\begin{aligned} \infty \sin\left(\frac{2}{\infty}\right) \\ \infty \sin 0 \\ \infty \cdot 0 \end{aligned}$$

Sol #1

$$\lim_{n \rightarrow \infty} \cancel{n} \frac{\sin\left(\frac{2}{\cancel{n}}\right)}{\frac{2}{\cancel{n}}}$$

$$= (1) 2 = 2$$

Sol #2

$$\lim_{n \rightarrow \infty} \frac{\sin\left(\frac{2}{n}\right)}{\frac{1}{n}} \cdot \frac{2}{2}$$

$$\lim_{n \rightarrow \infty} \frac{2 \sin\left(\frac{2}{n}\right)}{\frac{2}{n}}$$

$$= 2 (1) = 2$$

$$\lim_{n \rightarrow \infty} n \tan\left(\frac{\pi}{n}\right)$$

$$\begin{aligned} \infty \tan\left(\frac{\pi}{\infty}\right) \\ \infty \tan 0 \\ \infty \cdot 0 \end{aligned}$$

$$\lim_{n \rightarrow \infty} \cancel{n} \frac{\tan\left(\frac{\pi}{\cancel{n}}\right)}{\frac{\pi}{\cancel{n}}}$$

$$= (1) \pi = \pi$$



Tricky question !!



$$\lim_{n \rightarrow \infty} \left(\frac{\sin(n)}{n^2 \sin\left(\frac{1}{n}\right)} \right)$$

$$\frac{\sin \infty}{\infty \sin\left(\frac{1}{\infty}\right)}$$

Sandwich

$$\frac{\sin \infty}{\infty \sin 0}$$

قوانين

أوجله لشكل لقوانين
والى بقا
Sandwich

$$\lim_{n \rightarrow \infty}$$

$$\frac{\sin n}{n^2 \sin\left(\frac{1}{n}\right)}$$

$$\lim_{n \rightarrow \infty}$$

$$\frac{\sin n}{n \sin\left(\frac{1}{n}\right)}$$

Sandwich

$$\lim_{n \rightarrow \infty}$$

$$\frac{\frac{\sin n}{n}}{\frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}}} = \frac{0}{1} = 0$$

قوانين

$$\lim_{n \rightarrow \infty} \frac{\sin n}{n}$$

$$-1 \leq \sin n \leq 1$$

÷ n

$$\frac{-1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{-1}{n} = \frac{-1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = \frac{1}{\infty} = 0$$

⇒ ∴ By Sandwich Theorem

$$\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0$$

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$$\lim_{n \rightarrow \infty} a^n$$



$$\lim_{n \rightarrow \infty} a^n$$

$$\infty, \text{ if } a > 1$$

$$0, \text{ if } |a| < 1$$

Evaluate the limit if it exist

$$\lim_{n \rightarrow \infty} \left(\frac{5}{3}\right)^n$$

$$= \infty$$

Since $\frac{5}{3} > 1$

$$\lim_{n \rightarrow \infty} \left(\frac{4}{5}\right)^n$$

$$= 0$$

Since $|\frac{4}{5}| < 1$

$$\lim_{n \rightarrow \infty} \left(\frac{\pi}{e}\right)^n$$

$$= \infty$$

Since $\frac{\pi}{e} > 1$

$$\lim_{n \rightarrow \infty} \left(\frac{-5}{3}\right)^n$$

DNE

Since $-\frac{5}{3} < -1$

$$\lim_{n \rightarrow \infty} \left(\frac{-2}{3}\right)^n$$

$$= 0$$

Since $|\frac{-2}{3}| < 1$



$$\lim_{n \rightarrow \infty} \left(1 + \frac{a}{n} \right)^{kn} = e^{ak}$$



$$\lim_{n \rightarrow \infty} \left(1 + \frac{a}{bn + c} \right)^{kn} = e^{\frac{ak}{b}}$$

Evaluate the limit if it exist

$$\lim_{n \rightarrow \infty} \left(1 + \frac{5}{n} \right)^{3n}$$

$$= e^{5(3)} = e^{15}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{5}{2n + 4} \right)^{3n}$$

$$= e^{\frac{5(3)}{2}} = e^{\frac{15}{2}}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^{3n}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n}{n} + \frac{1}{n} \right)^{3n}$$

$$= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^{3n}$$

$$= e^{(1)(3)} = e^3$$

$$\lim_{n \rightarrow \infty} \left(\frac{n + \ln 2}{n} \right)^n$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n}{n} + \frac{\ln 2}{n} \right)^n$$

$$= \lim_{n \rightarrow \infty} \left(1 + \frac{\ln 2}{n} \right)^n$$

$$= e^{\ln 2} = 2$$

$$\lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^{2n}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n+1-1}{n+1} \right)^{2n}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n+1}{n+1} + \frac{-1}{n+1} \right)^{2n}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{-1}{n+1} \right)^{2n} = e^{(-1)(2)} = e^{-2}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n-4}{n+5} \right)^{2n}$$

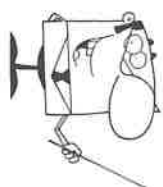
$$\lim_{n \rightarrow \infty} \left(\frac{n+5-5-4}{n+5} \right)^{2n}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n+5-9}{n+5} \right)^{2n}$$

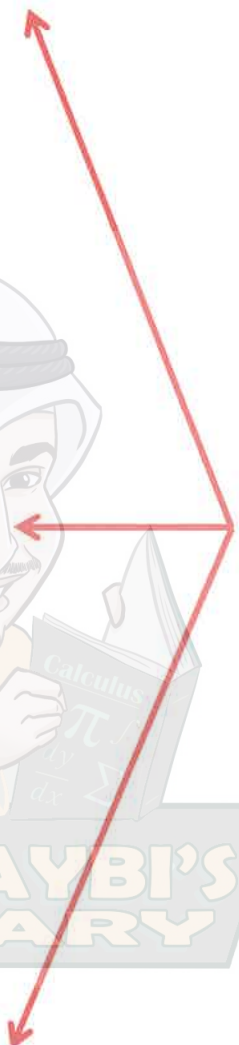
$$\lim_{n \rightarrow \infty} \left(\frac{\cancel{n+5} - 9}{\cancel{n+5}} + \frac{-9}{n+5} \right)^{2n}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{-9}{n+5} \right)^{2n} = e^{-9(2)} = e^{-18}$$

الإستنتاجات



عند فشل التعويض

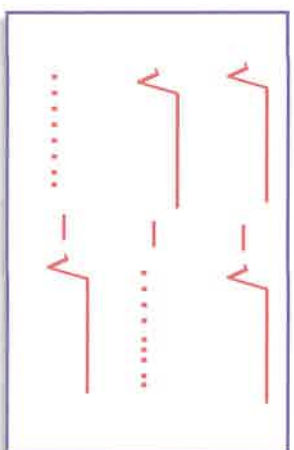


إذا عدي a^n بالبسط والمقام

يتم القسمة على أكبر a^n بالمقام

إذا عدي a^n بالبسط والمقام

1) أكبر عامل مشترك تحت الجذر
2) أكبر عامل مشترك بالبسط والمقام



يتم الضرب بالمرافق



إذا عني a^n بالبسط والمقام



يتم القسمة على أكبر a^n بالمقام



Evaluate the limit if it exist

$$\lim_{n \rightarrow \infty} \frac{3n^2 + 4^n}{5^n + n}$$

$$\frac{\infty + \infty}{\infty + \infty} = \frac{\infty}{\infty}$$

$\div 5^n$

$$\lim_{n \rightarrow \infty}$$

$$\frac{\frac{3n^2}{5^n} + \frac{4^n}{5^n}}{\frac{5^n}{5^n} + \frac{n}{5^n}}$$

$$\lim_{n \rightarrow \infty}$$

$$\frac{\frac{3n^2}{5^n} + \left(\frac{4}{5}\right)^n}{1 + \frac{n}{5^n}}$$

$$= \frac{0+0}{1+0} = 0$$

$$\lim_{n \rightarrow \infty} \frac{3n^2}{5^n} = \frac{\infty}{\infty}$$

Let $f(x) = \frac{3x^2}{5^x}$

$$\lim_{x \rightarrow \infty} \frac{3x^2}{5^x} = \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{6x}{5^x \ln 5} = \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{6}{5^x (\ln 5)^2} = \frac{0}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{n}{5^n} = \frac{\infty}{\infty}$$

Let $f(x) = \frac{x}{5^x}$

$$\lim_{x \rightarrow \infty} \frac{x}{5^x} = \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{1}{5^x \ln 5} = \frac{1}{\infty} = 0$$



Tricky question !!



$$\lim_{n \rightarrow \infty} \frac{3^{n+2} + 4^{n+1}}{3^n + 4^n}$$

8/8

$\lim_{n \rightarrow \infty}$

$$\frac{3^n \cdot 3^2 + 4^n \cdot 4}{3^n + 4^n}$$

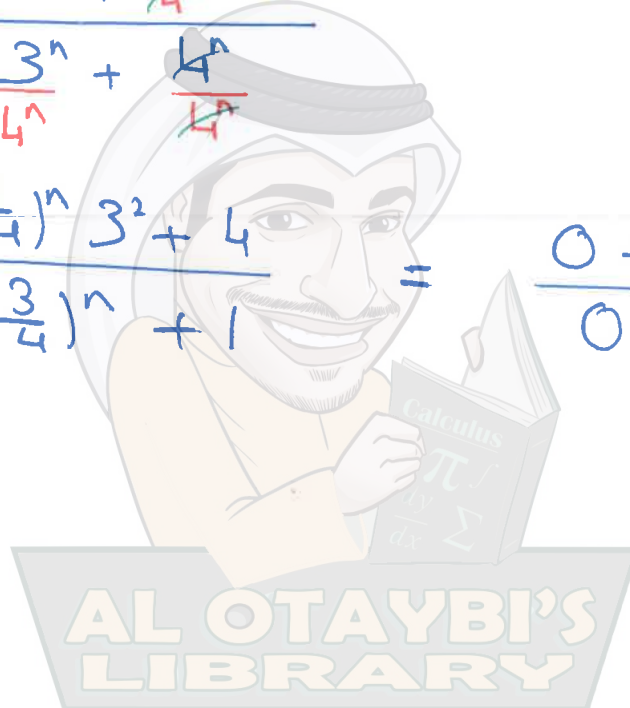
$\div 4^n$

$\lim_{n \rightarrow \infty}$

$$\frac{\frac{3^n}{4^n} 3^2 + \frac{4^n}{4^n} 4}{\frac{3^n}{4^n} + \frac{4^n}{4^n}}$$

$\lim_{n \rightarrow \infty}$

$$\frac{\left(\frac{3}{4}\right)^n 3^2 + 4}{\left(\frac{3}{4}\right)^n + 1} = \frac{0 + 4}{0 + 1} = 4$$



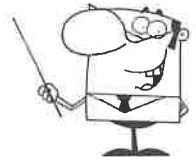


إذا عندي **حدودية** بالبسط والمقام



(1) أكبر عامل مشترك تحت الجذر

(2) أكبر عامل مشترك بالبسط والمقام



Evaluate the limit if it exist

$$\lim_{n \rightarrow \infty} \frac{5n^2 + 3n}{2n^2 + 4n + 5}$$

$$\frac{\infty}{\infty}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n^2} \left[5 + \frac{3}{n} \right]}{\cancel{n^2} \left[2 + \frac{4}{n} + \frac{5}{n^2} \right]} = \frac{5 + \frac{3}{\infty}}{2 + \frac{4}{\infty} + \frac{5}{\infty}} = \frac{5}{2}$$

$$\lim_{n \rightarrow \infty} \frac{3n + 1}{\sqrt{n^2 + 1}}$$

$$\frac{\infty}{\infty}$$

$$\lim_{n \rightarrow \infty} \frac{3n + 1}{\sqrt{n^2 \left(1 + \frac{1}{n^2} \right)}}$$

$$\lim_{n \rightarrow \infty} \frac{3n + 1}{n \sqrt{1 + \frac{1}{n^2}}}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n} \left[3 + \frac{1}{n} \right]}{\cancel{n} \sqrt{1 + \frac{1}{n^2}}} = \frac{3 + 0}{\sqrt{1 + 0}} = 3$$

$$\lim_{n \rightarrow \infty} \frac{n+5}{5n + \ln n}$$

$$\frac{\infty}{\infty + \infty} \quad \frac{\infty}{\infty}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n} \left[1 + \frac{5}{n} \right]}{\cancel{n} \left[5 + \frac{\ln n}{n} \right]} = \frac{1+0}{5+0} = \frac{1}{5}$$

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} \quad \frac{0}{0}$$

$$\text{Let } f(x) = \frac{\ln x}{x}$$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

$$\text{LiHop} \quad \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = \frac{1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{n^{10} + 5n}{n^{10} + e^n}$$

$$\frac{\infty + \infty}{\infty + \infty} \quad \frac{\infty}{\infty}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n^{10}} \left[1 + \frac{5n}{n^{10}} \right]}{\cancel{n^{10}} \left[1 + \frac{e^n}{n^{10}} \right]}$$

$$\lim_{n \rightarrow \infty} \frac{1 + \frac{5}{n^9}}{1 + \frac{e^n}{n^{10}}} = \frac{1+0}{1+\infty} = \frac{1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{e^n}{n^{10}} \quad \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^{10}} = \infty$$

(after applying LiHop Rule 10 times)



Tricky Question !!



$$\lim_{n \rightarrow \infty} \frac{n^3 + 5n}{5n^3 + \cos(n) + n^{-2}}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n^3} \left[1 + \frac{5}{n^2} \right]}{\cancel{n^3} \left[5 + \frac{\cos n}{n^3} + \frac{n^{-2}}{n^3} \right]}$$

$$\lim_{n \rightarrow \infty} \frac{1 + \frac{5}{n^2}}{5 + \frac{\cos n}{n^3} + \frac{1}{n^5}} = \frac{1+0}{5+0+0} = \frac{1}{5}$$

$$\lim_{n \rightarrow \infty} \frac{\cos n}{n^3}$$

$$-1 \leq \cos n \leq 1$$

$$\div n^3$$

$$\frac{-1}{n^3} \leq \frac{\cos n}{n^3} \leq \frac{1}{n^3}$$

$$\lim_{n \rightarrow \infty} \frac{-1}{n^3} = \frac{-1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^3} = \frac{1}{\infty} = 0$$

∴ By Sandwich Theorem

$$\lim_{n \rightarrow \infty} \frac{\cos n}{n^3} = 0$$



عند فشل التعويض وعندي

$$\begin{array}{rcl} \sqrt{\quad} & - & \sqrt{\quad} \\ \sqrt{\quad} & - & \dots\dots\dots \\ \dots\dots\dots & - & \sqrt{\quad} \end{array}$$



→ يتم الضرب بالمرافق ←

Evaluate the limit if it exist

$$\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n+5})$$

$\infty - \infty$

$$\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n+5}) \cdot \frac{\sqrt{n+1} + \sqrt{n+5}}{\sqrt{n+1} + \sqrt{n+5}}$$

$$\lim_{n \rightarrow \infty} \frac{n+1 - (n+5)}{\sqrt{n+1} + \sqrt{n+5}}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n}+1-\cancel{n}-5}{\sqrt{n+1} + \sqrt{n+5}} = \frac{-4}{\infty + \infty} = \frac{-4}{\infty} = 0$$



$$\lim_{n \rightarrow \infty} (-1)^n f(n)$$



$$\lim_{n \rightarrow \infty} |(-1)^n f(n)|$$

النتيجة if

$$= 0$$



$$\lim_{n \rightarrow \infty} (-1)^n f(n) = 0$$

$$\text{رقم} \\ \pm \infty$$



$$\lim_{n \rightarrow \infty} (-1)^n f(n) \text{ DNE}$$

Evaluate the limit if it exist

$$\lim_{n \rightarrow \infty} (-1)^n \frac{n^3}{n^3 + 2n^2 + 1}$$

$$\lim_{n \rightarrow \infty} \left| (-1)^n \frac{n^3}{n^3 + 2n^2 + 1} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n^3}{n^3 + 2n^2 + 1} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n^3}{n^3 (1 + \frac{2}{n} + \frac{1}{n^3})} \right| = \left| \frac{1}{1+0+0} \right| = \frac{1}{1} = 1$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} (-1)^n \frac{n^3}{n^3 + 2n^2 + 1} \text{ DNE}$$

هناك حل آخر

$$\lim_{n \rightarrow \infty} \frac{(-1)^n \tan^{-1} n}{1 + \ln n}$$

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^n \tan^{-1} n}{1 + \ln n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{\tan^{-1} n}{1 + \ln n} \right| = \left| \frac{\frac{\pi}{2}}{\infty} \right| = 0$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} \frac{(-1)^n \tan^{-1} n}{1 + \ln n} = 0$$

$$\lim_{n \rightarrow \infty} (-1)^n n \sin \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \left| (-1)^n n \sin \frac{1}{n} \right|$$

$$\lim_{n \rightarrow \infty} \left| n \sin \frac{1}{n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \cancel{n} \frac{\sin \frac{1}{n}}{\frac{1}{n}} \cancel{\frac{1}{n}} \right| = 1 = 1$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} (-1)^n n \sin \frac{1}{n} \text{ DNE}$$

$$\lim_{n \rightarrow \infty} \left[\cos \frac{1}{n} + (-1)^n \frac{2n^2 + 1}{n^2 + 5} \tan^{-1} n \right]$$

$$\lim_{n \rightarrow \infty} \cos \frac{1}{n} + \lim_{n \rightarrow \infty} (-1)^n \frac{2n^2 + 1}{n^2 + 5} \tan^{-1} n$$

$$= \cos \frac{1}{\infty}$$

$$= \cos 0$$

$$= 1$$

$$\lim_{n \rightarrow \infty} \left| (-1)^n \frac{2n^2 + 1}{n^2 + 5} \tan^{-1} n \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{2n^2 + 1}{n^2 + 5} \tan^{-1} n \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n^2 (2 + \frac{1}{n^2})}{n^2 (1 + \frac{5}{n^2})} \tan^{-1} n \right|$$

$$= \left| \frac{2+0}{1+0} \frac{\pi}{2} \right|$$

AL OTAYBI'S LIBRARY $\neq \pi$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} (-1)^n \frac{2n^2 + 1}{n^2 + 5} \tan^{-1} n \text{ DNE}$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} \left[\cos \frac{1}{n} + (-1)^n \frac{2n^2 + 1}{n^2 + 5} \tan^{-1} n \right]$$

DNE



Factorial !



واحد !

أكثر من !



Sandwich
Theorem



أبطل الكبير لحين
شطب الصغير

$$0! = 1$$

$$1! = 1$$

$$2! = (2)(1) = 2$$

$$3! = (3)(2)(1) = 6$$

$$6! = (6)(5)(4)(3)(2)(1)$$

$$n! = n(n-1)(n-2) \dots (3)(2)(1) \\ = n(n-1)!$$

Evaluate the limit if it exist

$$\lim_{n \rightarrow \infty} \frac{(3n-2)!}{(3n+2)!}$$

عند وجود ! بالمت
← نبتل الأكبر →
لا مكانية الامتصار
للمت

$$\lim_{n \rightarrow \infty}$$

$$\frac{(3n-2)!}{(3n+2)(3n+1)(3n)(3n-1)(3n-2)!} = \frac{1}{\infty} = 0$$



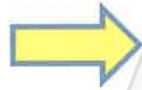
Relations



$$\frac{n!}{n^n}$$

$$= \frac{\cancel{n} (n-1)(n-2) \dots (3)(2)(1)}{\cancel{n} n n \dots n n n}$$

$$= \left(\frac{n-1}{n}\right) \left(\frac{n-2}{n}\right) \dots \left(\frac{3}{n}\right) \left(\frac{2}{n}\right) \left(\frac{1}{n}\right)$$



$$0 < \frac{n!}{n^n} < \frac{2}{n}$$



$$\frac{2^n}{n!}$$

$$= \frac{(2)(2)(2) \dots (2)(2)(2)}{n(n-1)(n-2) \dots (3)(2)(1)}$$

$$= \left(\frac{2}{n}\right) \left(\frac{2}{n-1}\right) \left(\frac{2}{n-2}\right) \dots \left(\frac{2}{3}\right) (2)$$



$$0 < \frac{2^n}{n!} < \frac{2}{n} 2$$



Evaluate the limit if it exist

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n}$$

$$\frac{n!}{n^n} = \frac{\cancel{n} (n-1) (n-2) \dots (3) (2) (1)}{\cancel{n} n n \dots n n n}$$

$$0 < \frac{n!}{n^n} < \frac{2}{n}$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{2}{n} = \frac{2}{\infty} = 0$$

$\Rightarrow \therefore$ By Sandwich Theorem $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$

$$\lim_{n \rightarrow \infty} \frac{2^n}{n!}$$

$$\frac{2^n}{n!} = \frac{(2)(2)(2) \dots (2)(2)(2)}{n(n-1)(n-2) \dots (3)(2)(1)}$$

$$0 < \frac{2^n}{n!} < \frac{2}{n} \quad (2)$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{2}{n} = \frac{2}{\infty} = 0$$

$\Rightarrow \therefore$ By Sandwich Theorem $\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0$

خطوات حل مسائل اللمت

التعويض المباشر

$$0^0 \text{ أو } \infty^0 \text{ أو } 1^\infty$$

$$0 \cdot \infty$$

$$\infty - \infty$$

$$\frac{0}{0} \text{ أو } \frac{\infty}{\infty}$$

$$\infty \text{ رقم أو } \infty$$



$$1) y = \dots\dots\dots$$

$$2) \ln y = \ln \dots\dots\dots$$

أخذ اللمت للطرف اليمين

النتائج =

Final Answer

e

أخذي دالة
مقام للتأنيدي

توحيد
المقامات

تطبيق
L'Hospital
Rule

Stop !!

Note

إذا عندي

e

ln

هي المقام

هي البسط



Trick



جزء من التعويض شكل

اللوبيتال

يتم تطبيق اللوبيتال على هذا

الجزء لـ

بدون عوار

راس

معرفة أنا

وين

Do Not Forget !!

$$\frac{0}{\text{عدد}} = 0, \quad \frac{\text{عدد}}{\infty} = 0, \quad \frac{\text{عدد}}{0} = \infty$$